

On the Complexity of Safety Checking in regular Open Multi Agent Systems

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May 2025

1 Introduction

2 OMAS [4] & Regular OMAS [5]

A Multi agent system in which agents may enter or leave the system at runtime is known as *open MAS*. In this section, we review the syntax and semantics of Open Interpreted System (OIS), proposed by [4]. OIS is based on interpreted systems, originally proposed by [3] to provide computationally grounded semantics to model temporal and epistemic properties of multi-agent and distributed computer systems. Interpreted systems are defined over a set of agents $\{1, 2, \dots, n\}$ with local states for each agent and the environment. A snapshot of the system is represented using a *global state* containing local states of each agent and the environment. Further, the evolution of the system over time can be modelled by the notion of *run*.

2.1 Syntax and Semantics of Open MAS

An agent template defines the behaviour of individual agents in the open MAS. This is given as follows.

Definition 1 (Agent). Agent template is a tuple $AT = \langle L, \iota, Act, P, tr \rangle$ where:

- $L \neq \emptyset$ is the set of local states of the agents.
- $\iota \in L$ is the unique initial local state of agents.
- Act is the non-empty set of actions the agents executes. Act contains a special symbol τ that corresponds to “no action”.
- $P : L \longrightarrow \mathcal{P}(Act)$ is a protocol function that gives the actions enabled at the current state for the agents. Note that $\tau \in P(l)$, for every $l \in L$.
- $tr : L \times Act \times \mathcal{P}(Act) \times Act_E \longrightarrow L$ is the local transition function returning the next state of the agent from its current state based on the action executed by that agent, set of actions executed by other agents in the system and the action executed by the environment. Note, that for any $l \in L$, $tr(l, \tau, A, b) = l$, where $A \subseteq Act$ and $b \in Act_E$.

The environment behaviour is defined as follows.

Definition 2 (Environment). The environment is a tuple $E = \langle L_E, \iota_E, Act_E, P_E, tr_E \rangle$ where:

- $L_E \neq \emptyset$ is the set of local states of the environment and $L \cap L_E = \emptyset$.
- $\iota_E \in L_E$ is the unique initial state.
- $Act_E \neq \emptyset$ is a set of actions that the environment executes.
- $P_E : L_E \longrightarrow \mathcal{P}(Act_E)$ is a protocol function that defines which actions are enabled at a given local state in the environment.
- $tr_E : L_E \times Act_E \times \mathcal{P}(Act) \longrightarrow L_E$ is the transition function that gives the next state of the environment given its current state, the action it executes and the set of actions executed by the agents.

We combine the agent templates and environment to obtain open MAS as follows.

Definition 3 (OIS). An open interpreted MAS (OIS) is a pair $\mathcal{O} = \langle AT, E \rangle$ where:

- AT is the agent template and
- E is the environment as defined above.

Now we define the semantics of open MAS. In any instance, the system (open MAS) comprises the environment and zero or more agents. Each agent is assigned an integer – its identity – which is unique to that particular agent. The identity is assigned when the agent joins the system, and it will not change until the agent leaves the system. The environment state along with the agent states and their corresponding identities, at any point of time, gives the snapshot (global state) of the system.

When n agents are present in the system, the set of tuples that characterize it at a particular instance of time is denoted by G_n where G_n is a set of $n + 1$ tuples of the form $(L \times \mathbb{Z}^+) \times \dots \times (L \times \mathbb{Z}^+) \times L_E$ that gives the set of global state of the system of size n . For a global state g , $g.j$ gives the local state of j^{th} agent and $id(g.j)$ give the identity of j^{th} agent, whereas $g.E$ gives the state of the environment. $G \triangleq \bigcup_{n \in \mathbb{N}} G_n$ is the set of all global states of any size.

For a given $n \in \mathbb{N}$, the set of joint actions is $ACT_n = Act^n \times Act_E$. The global transition relation R_n , for a given n , depends on the projection of joint action into a set. A joint action in ACT_n is denoted by $\mathbf{a}$ and the projection of joint action into a set as $\dot{\mathbf{a}} \triangleq \{\mathbf{a}.j \mid \mathbf{a}.j \text{ is the action of agent } j \text{ in the joint action } \mathbf{a} \text{ and } 1 \leq j \leq n\}$. For a given $i \in [n]$, we define $\dot{\mathbf{a}}_{-i}$ – the set of actions in $\mathbf{a}$ except the action of agent i – as follows: $\dot{\mathbf{a}}_{-i} \triangleq \{\mathbf{a}.j \mid 1 \leq j < i\} \cup \{\mathbf{a}.j \mid i < j \leq n\}$. Note, $\dot{\mathbf{a}}_{-i}$ may include $\mathbf{a}.i$.

We now define the temporal evolution of the global states. First, we define the global transition relation of a fixed-size open MAS. Then, we extend the global transition relation for the open MAS with agents leaving and joining the system at runtime.

Definition 4 (The global transition relation of size n). $R_n \subseteq G_n \times G_n$ on a set G_n of global states is defined by $(g, g') \in R_n$ iff there is some $\mathbf{a} \in ACT_n$ such that the following hold:

- $\mathbf{a}.E \in P_E(g.E)$ and for all $j \in [n]$, $\mathbf{a}.j \in P(g.j)$ where $\mathbf{a}.E$ is the environment action and P is the protocol function for the agent.
- $g'.E = tr_E(g.E, \mathbf{a}.E, \dot{\mathbf{a}})$
- for all $j \in [n]$ it is the case that $id(g'.j) = id(g.j)$ and $g'.j = tr(g.j, \mathbf{a}.j, \dot{\mathbf{a}}_{-j}, \mathbf{a}.E)$, for the j th agent in the system.

The first condition ensures that the action of both environment and agents are enabled as per the protocol functions of the environment and agents, respectively. The second condition guarantees that the transition of the environment is consistent with respect to t_E . The last condition ensures that the agent's identity remains the same after the global transition and each agent's transition is consistent with respect to the corresponding transition relation t_i .

Definition 5 (Global transition relation). The global transition relation $R \subseteq G \times G$ on the set G of global states is defined by $(g, g') \in R$ iff any of the following conditions are met:

Joint action: There is $n \in \mathbb{N}$ such that $(g, g') \in R_n$. This corresponds to a global transition of the system of size n , already defined above.

Agent joining: (i) There is $n \in \mathbb{N}$ such that $g \in G_n$ and $g' \in G_{n+1}$. Also, let $\mathbb{D}$ be the set of agent identities that occur in g . (ii) For all $j \in [n]$ it is the fact that $g.j = g'.j$ and $g.E = g'.E$. (iii) $g'.(n+1) = \iota$ where ι is the initial local state of the agent and $id(g'.(n+1)) \in \mathbb{N} \setminus \mathbb{D}$.

Agent leaving: (i) There is $n \in \mathbb{N}$ such that $g \in G_n$ and $g' \in G_{n-1}$; (ii) There is $\kappa \in [n]$ such that $g' = g_{-\kappa}$, where $g_{-\kappa}$ denotes the tuple resulting from removing the κ th component from the tuple g .

Now, a model of OIS may be defined.

Definition 6 (Model). The associated model of an OIS $\mathcal{O} = \langle AT, E \rangle$ is a tuple $O = \langle G, g_0, R \rangle$ where G is the set of global states reachable via R from the initial global state $g_0 = \langle \iota_E \rangle$. Also, let $Runs(\mathcal{O})$ be the set of all runs of O starting from the initial global state g_0 .

2.2 Regular OMAS

A subclass of open MAS called Regular OMAS or **Rutwiya Systems** is defined in this section. Later, in Section 4, we show that safety checking in this newly defined class is decidable in P-time.

In Rutwiya Systems, agents can leave the system at the time of agent leaving transition, but only from a fixed **leaving state**. We can modify the agent template as follows: $AT = \langle L, \iota, Act, P, tr, leave \rangle$ where $leave \in L$ is the leaving state. Also, we insist that $P(leave) = \emptyset$, that is, no action is possible from the leave state. Hence, for the global transition relation R , agent leaving can be redefined as:

Agent leaving: (i) There is $n \in \mathbb{N}$ such that $g \in G_n$ and $g' \in G_{n-1}$; (ii) There is $\kappa \in [n]$ such that $g.\kappa = leave$ and $g' = g_{-\kappa}$, where $g_{-\kappa}$ denotes the tuple resulting from removing the κ th component from the tuple g .

3 Reconfigurable Broadcast Networks

We now define a model for the *leader* version of **Reconfigurable Broadcast Networks** (RBN) [2], inspired by the ASMS-leader of [1]. In order to specify an **RBN-leader**, we define the notion of **topology** (essentially, graphs) used to denote network **configurations** and **protocols** running on each node of the topology. An RBN-leader $\mathcal{R}$ consists of a pair of protocols $\langle \mathcal{R}_C, \mathcal{R}_L \rangle$ defined over a common finite alphabet Σ . $\mathcal{R}_C$ is called the **contributor protocol** and $\mathcal{R}_L$ is called the **leader protocol**. A topology γ may define a **configuration** of an RBN-leader $\mathcal{R}$ wherein there is exactly one node that runs the leader protocol, and all other nodes run the contributor protocol.

Definition 7. A **leader protocol** is a tuple $R_L = \langle Q_L, \delta_L, Q_L^0 \rangle$, where Q_L is a finite set of control states, $\delta_L \subseteq Q_L \times (\{\tau\} \cup \{!a, ?a | a \in \Sigma\}) \times Q_L$ is the transition relation, and $Q_L^0 \subseteq Q_L$ is a set of initial control states.

Definition 8. A **contributor protocol** is a tuple $R_C = \langle Q_C, \delta_C, Q_C^0 \rangle$, where Q_C is a finite set of control states, $\delta_C \subseteq Q_C \times (\{\tau\} \cup \{!b, ?b | b \in \Sigma\}) \times Q_C$ is the transition relation, and $Q_C^0 \subseteq Q_C$ is a set of initial control states.

The label τ denotes a local transition, $!a$ ($!b$) denotes broadcasting of message a (b) to all adjacent nodes, and the label $?a$ ($?b$) denotes the reception of message a (b) from some adjacent node. For $q \in Q_L$ and $a \in \Sigma$ we define the set $\delta_L^a(q) = \{q' \in Q_L | \langle q, ?a, q' \rangle \in \delta_L\}$ of sink/target states of q for the reception of message a in the leader protocol. For $q \in Q_C$ and $b \in \Sigma$ we define the set $\delta_C^b(q) = \{q' \in Q_C | \langle q, ?b, q' \rangle \in \delta_C\}$ of sink/target states of q for the reception of message b in the contributor protocol.

Now we define the notion of topology γ for an RBN-leader $\mathcal{R} = \langle \mathcal{R}_C, \mathcal{R}_L \rangle$ as follows.

Definition 9. A topology is a labelled undirected graph $\gamma = \langle V, E, \mathcal{L} \rangle$, where

- V is a finite set of nodes with a designated leader node v_L ,
- $E \subseteq (V \times V) \setminus \{\langle v, v \rangle | v \in V\}$ is a finite set of edges, and
- $\mathcal{L}$ is a labelling function $\mathcal{L} : V \longrightarrow (Q_C \cup Q_L)$ for nodes in the topology such that $\mathcal{L}(v_L) \in Q_L$ and $\forall v \in V, \mathcal{L}(v) \in Q_C$.

We write $u \sim_\gamma v$ when the node u is adjacent to v in the graph γ , that is, when $\langle u, v \rangle \in E$. Also, let $\mathcal{L}(\gamma)$ be the set of labels appearing in nodes of γ :

$$\mathcal{L}(\gamma) = \{q \in Q_C \cup Q_L \mid \exists v \in V : \mathcal{L}(v) = q\}.$$

Given an RBN-leader $\mathcal{R} = \langle R_L, R_C \rangle$, a **configuration** of $\mathcal{R}$ is a graph (topology) γ as already mentioned. A configuration is said to be initial if $\mathcal{L}(\gamma) \subseteq Q_L^0 \cup Q_C^0$. We use $\mathcal{C}$ to denote the set of all configurations in $\mathcal{R}$ and $\mathcal{C}_0$ to denote the set of initial configurations in $\mathcal{R}$.

Given an RBN-leader $\mathcal{R} = \langle R_L, R_C \rangle$, a reconfigurable broadcast network is defined by the transition system $RBN(\mathcal{R}) = \langle \mathcal{C}, \longrightarrow, \mathcal{C}_0 \rangle$ where the transition relation $\longrightarrow \subseteq \mathcal{C} \times \mathcal{C}$ is such that: for any pair of configurations $\gamma, \gamma' \in \mathcal{C}$ with $\gamma = \langle V, E, \mathcal{L} \rangle$ and $\gamma' = \langle V, E', \mathcal{L}' \rangle$ we have $\gamma \longrightarrow \gamma'$ iff one of the following conditions holds:

τ transition Either a single contributor makes a τ transition – $\exists v \in V$ such that $(\mathcal{L}(v), \tau, \mathcal{L}'(v)) \in \delta_C$ and $\forall v' \in (V - \{v\}) \mathcal{L}(v') = \mathcal{L}'(v')$, also, $\mathcal{L}(v_L) = \mathcal{L}'(v_L)$ or the leader makes a τ transition – $(\mathcal{L}(v_L), \tau, \mathcal{L}'(v_L)) \in \delta_L$ and $\forall v' \in V \mathcal{L}(v') = \mathcal{L}'(v')$. In both cases $E' = E$.

broadcast $E' = E$ (topology does not change) and

leader broadcasts: for some $a \in \Sigma$, $\langle \mathcal{L}(v_L), !a, \mathcal{L}'(v_L) \rangle \in \delta_L$ and for all $u \in V - \{v_L\} : v_L \sim_\gamma u$, $\langle \mathcal{L}(u), ?a, \mathcal{L}'(u) \rangle \in \delta_C$ **OR**

contributor broadcasts: for some $b \in \Sigma$ and $v \in V - \{v_L\}$, $\langle \mathcal{L}(v), !b, \mathcal{L}'(v) \rangle \in \delta_C$ and for all $u \in V - \{v_L\} : v \sim_\gamma u$, $\langle \mathcal{L}(u), ?b, \mathcal{L}'(u) \rangle \in \delta_C$ and if $v \sim_\gamma v_L$ then $\langle \mathcal{L}(v_L), ?b, \mathcal{L}'(v_L) \rangle \in \delta_L$.

graph reconfiguration: Topology changes – $E' \subseteq V \times V \setminus \{\langle v, v \rangle \mid v \in V\}$ and $\mathcal{L} = \mathcal{L}'$ (states – labels – do not change).

We use $\longrightarrow^*$ to denote the reflexive and transitive closure of $\longrightarrow$. We use one of the parameterized reachability problems [2] of RBN-leader to show that safety checking problem in (regular) OMAS is PTIME-solvable. The following text on parameterized reachability problems is taken from [2].

3.1 Parameterized Reachability Problems

Given a RBN-leader $\mathcal{R}$ a cardinality constraint φ over $\mathcal{R}$ is a formula which defines lower and upper bounds for the number of occurrences of each control state in a configuration. The formulae are defined by the following grammar, where $a \in \mathbb{N}$, $q \in (Q_L \cup Q_C)$, and $b \in (\mathbb{N} - \{0\}) \cup \{+\infty\}$:

$$\varphi ::= a \leq \#q < b \mid \varphi \wedge \varphi \mid \varphi \vee \varphi \mid \neg \varphi$$

We denote by CC the class of cardinality constraints, by $CC[\geq 1]$ the class in which negation is forbidden and atomic proposition have only the form $\#q \geq 1$ (there exists at least one occurrence of q), and finally by $CC[\geq 1, = 0]$ the class of cardinality constraints as in $CC[\geq 1]$ but where atoms can also be of the form $\#q = 0$.

Given a configuration $\gamma = \langle V, E, \mathcal{L} \rangle$ of $\mathcal{R}$ and $q \in Q_L \cup Q_C$, we denote by $\# \gamma(q)$ the number of vertices in γ labeled by q , that is $\# \gamma(q) = |\{v \in V \mid \mathcal{L}(v) = q\}|$. The satisfaction relation $\models$ for atomic formulas is defined as follows: $\gamma \models a \leq \#q < b$ iff $a \leq \# \gamma(q) < b$. $\models$ can be extended to compound formulas in a natural way using structural induction. We are now ready to state the **cardinality reachability problem** (CRP) for RBN-leader:

Input: An RBN-leader $\mathcal{R}$ with $RBN(\mathcal{R}) = \langle \mathcal{C}, \longrightarrow, \mathcal{C}_0 \rangle$ and a cardinality constraint φ .

Output: Yes, if $\exists \gamma_0 \in \mathcal{C}_0$ and $\gamma_1 \in \mathcal{C}$ such that $\gamma_0 \longrightarrow^* \gamma_1$ and $\gamma_1 \models \varphi$; no, otherwise.

Lemma 10 ([2]). *CRP for RBN-leader restricted to $CC[\geq 1]$ is PTime-complete.*

4 Encoding Regular OMAS into RBN-leader

Given a Regular OMAS $\mathcal{O} = \langle AT, E \rangle$, where

- $AT = \langle L, \iota, Act, P, tr, leave \rangle$ is the **agent template** and
- $E = \langle L_E, \iota_E, Act_E, P_E, tr_E \rangle$ is the **environment**,

we define an encoding of $\mathcal{O}$ as a RBN *with leader* $\mathcal{R}_{\mathcal{O}} = \langle \mathcal{R}_C, \mathcal{R}_L \rangle$, where

- $\mathcal{R}_C = (Q_C, \Sigma, \delta_C)$ is the **contributor protocol** and
- $\mathcal{R}_L = (Q_L, \Sigma, \delta_L)$ is the **leader protocol**,

that captures the behaviour of $\mathcal{O}$.

First, we initialize the components of $\mathcal{R}_C$ and $\mathcal{R}_L$ as follows:

- $Q_C = L \cup \{sleep\}$ and $Q_L = L_E$.
- $Q_C^0 = \{sleep\}$ and $Q_L^0 = \{\iota_E\}$.
- $\Sigma = \{\tau\}$ and $\Sigma = \emptyset$.
- $\delta_C = \{(sleep, \tau, \iota), (leave, \tau, sleep)\}$ and $\delta_L = \emptyset$. Clearly, $(sleep, \tau, \iota)$ corresponds to *agent joining* and $(leave, \tau, sleep)$ corresponds to *agent leaving*. Initially, all agents (contributors) are in the ‘sleep’ state and the leader in ι_E state.

Now for every environment transition tuple $\mathbf{tr} = (l_e, b, A, l'_e)$ in tr_E we augment $\mathcal{R}_C = (Q_C, \Sigma, \delta_C)$ and $\mathcal{R}_L = (Q_L, \Sigma, \delta_L)$ as follows.

Let $A = \{a_1, a_2, \dots, a_k\} \subseteq Act$ be the set of all agent actions executed in a joint action corresponding to the aforementioned environment transition, $\mathbf{tr}$. For $\mathbf{tr}$ we define a set of transition tuples for contributor and a set of transition tuples for leader which together describe all possible global joint transitions in $\mathcal{O}$ consistent with $\mathbf{tr}$.

For every transition $(l_1, a_1, A_1, l'_1) \in tr$, add a tuple $(l_1, ?(\langle a_1, \emptyset \rangle, b), l'_1)$ to δ_C

Similarly, for every transition $(l_2, a_2, A_2, l'_2) \in tr$, add a tuple $(l_2, ?(\langle a_2, \{a_1\} \rangle, b), l'_2)$ to δ_C

In general, for every transition $(l_i, a_i, A_i, l'_i) \in tr$, add a tuple $(l_i, ?(\langle a_i, \{a_1, a_2, \dots, a_{i-1}\} \rangle, b), l'_i)$ to δ_C

Correspondingly, we add the following set of transitions to δ_L where $l_1^e, l_2^e, \dots, l_{k-1}^e$ are freshly generated symbols not in L_E :

$$\{(l, !(\langle a_1, \emptyset \rangle, b), l_1^e), (l, !(\langle a_2, \{a_1\} \rangle, b), l_2^e), \dots, (l, !(\langle a_k, \{a_1, a_2, \dots, a_{k-1}\} \rangle, b), l_{k-1}^e)\}$$

Further, we update Σ and Σ as follows:

$$\Sigma = \Sigma \cup \{?(\langle a_1, \emptyset \rangle, b), \dots, ?(\langle a_k, \{a_1, a_2, \dots, a_{k-1}\} \rangle, b)\} \text{ and}$$

$$\Sigma = \Sigma \cup \{!(\langle a_1, \emptyset \rangle, b), \dots, !(\langle a_k, \{a_1, a_2, \dots, a_{k-1}\} \rangle, b)\}.$$

Also, we update Q_L as follows:

$$Q_L = Q_L \cup \{l_1^e, l_2^e, \dots, l_{k-1}^e\}.$$

Here we are making an order for the actions, means first perform action a_1 , then a_2 and so on. No other order will encourage.

Let $(G, \implies, g_0)$ be the semantics of $\mathcal{O}$ and $(\mathcal{C}, \longrightarrow, \mathcal{C}_0)$ be the semantics of $\mathcal{R}_{\mathcal{O}}$. We define a notion of **consistency** between the the global states in G and configurations in $\mathcal{C}$. First, let us define a set of *possibly consistent* configurations $\mathcal{C}_{\mathfrak{g}} \subseteq \mathcal{C}$. For any $\gamma = (V, E, \mathcal{L}) \in \mathcal{C}$ if $\mathcal{L}(v_L) \in L_E$ then $\gamma \in \mathcal{C}_{\mathfrak{g}}$.

We define two functions f_1 and f_2 in order to decide whether a given $g \in G$ and $\gamma \in \mathcal{C}_{\mathfrak{g}}$ are consistent with each other.

$f_1 : G \times (L \cup L_E) \longrightarrow \mathbb{N}$ such that

- for any $l \in L$, $f_1(g, l)$ is the number of times l occurs in g and
- for any $l_e \in L_E$, $f_1(g, l_e) = 1$ if $g.E = l_e$ and $f_1(g, l_e) = 0$, otherwise.

$f_2 : \mathcal{C}_{\mathfrak{g}} \times (L \cup L_E) \longrightarrow \mathbb{N}$ such that

- for any $l \in (L \cup L_E)$, $f_2(\gamma, l)$ is the number of nodes labelled by l and

Now, for any $g \in G$ and $\gamma \in \mathcal{C}_{\mathfrak{g}}$ we say that g and γ are consistent with each other if for all $l \in (L \cup L_E)$, $f_1(g, l) = f_2(\gamma, l)$. Note that for any $\gamma_0 \in \mathcal{C}_0$, γ_0 is consistent with g_0 .

Lemma 11 (Completeness). *For any $g \in G$ if $g_0 \implies^* g$ then there exists $\gamma_0 \in \mathcal{C}_0$ and $\gamma \in \mathcal{C}$ such that $\gamma_0 \longrightarrow^* \gamma$, where γ is consistent with g , respectively.*

Proof. Can be proved by induction on the length of the run $g_0 \implies^* g$.

Let $\rho = g^0 g^1 g^2 \dots g^i \dots g^n$ be the run of $\mathcal{O}$, where $g^0 = g_0$ and $g^n = g$. For every $0 \leq i \leq n$, we map g^i to a sequence of configurations in $\mathcal{R}_{\mathcal{O}}$, $\gamma^i \in \mathcal{O}_{\mathcal{O}}$. The concatenation of all such γ^i 's gives us a legal run $\gamma^0 \cdot \gamma^1 \cdot \gamma^2 \dots \gamma^i \dots \gamma^n = \gamma \in \text{Runs}(\mathcal{R}_{\mathcal{O}})$.

We construct γ inductively:

$n = 0$: $\gamma^0 = \gamma_0 = (V, E, \mathcal{L})$ where $\mathcal{L}(\gamma_0) \subseteq Q_L^0 \cup Q_C^0$. For all $l \in L$, $f_1(g, l) = f_2(\gamma, l) = 0$ and $f_1(g, l) = f_2(\gamma, l) = 1$ if $l = \iota_E$.

$n > 0$: Assume that γ^i is well defined for all $i < n$. Suppose σ^* is the last configuration in the sequence γ^{n-1} . We need to do the following, for some $\mathfrak{m}_n$ that we compute later: define $\gamma^n = \sigma_1 \sigma_2 \dots \sigma_{\mathfrak{m}_n}$ such that $\sigma^* \Rightarrow \sigma_1$ and for all $j \in \{1, 2, \dots, \mathfrak{m}_n - 1\}$, $\sigma_j \Rightarrow \sigma_{j+1}$.

As mentioned above, we define γ^n as a sequence of configurations $\sigma_1 \sigma_2 \dots \sigma_{\mathfrak{m}_n}$ where for all $j \in \{1, 2, \dots, \mathfrak{m}_n\}$, $\sigma_j = (V, E, \mathcal{L})$. We consider the three cases depending on the type of the transition (g^{n-1}, g^n) and define $(V, E, \mathcal{L})$ for all $j \in \{1, 2, \dots, \mathfrak{m}_n\}$:

- (g^{n-1}, g^n) is an **agents joining transition**: Let r' be the number of times ι occurs in g^{n-1} and r'' be the number of times ι occurs in g^n . Let $r = r'' - r'$, $\sigma^* = (V, E, \mathcal{L})$ and $\sigma_{\mathfrak{m}_n} = (V', E', \mathcal{L}')$. $V' = V - (r \cdot \text{sleep}) + (r \cdot \iota)$. $\forall v \in V' - V, \mathcal{L}(v) = \iota$.

- (g^{n-1}, g^n) is an **agents leaving transition**: Let r' be the number of times *leave* occurs in g^{n-1} and r'' be the number of times *leave* occurs in g^n . Let $r = r' - r''$, $\sigma^* = (V, E, \mathcal{L})$ and $\sigma_{m_n} = (V', E', \mathcal{L}')$. $V' = V - (r$

□

Lemma 12 (Soundness). *For every $\gamma = (V, E, \mathcal{L}) \in \mathcal{C}$ such that $\mathcal{L}(v_L) \in L_E$, if γ is reachable from some $\gamma_0 \in \mathcal{C}_0$ then there is a $g \in G$ such that g is reachable from g_0 and γ is consistent with g .*

Proof. We sketch the proof of soundness here. Suppose γ is reachable from some $\gamma_0 \in \mathcal{C}_0$ by a run $\rho = \gamma_0 \longrightarrow \gamma_1 \longrightarrow \gamma_2 \longrightarrow \dots \longrightarrow \gamma_\kappa = \gamma$. Each transition in ρ is either a reconfiguration (which we ignore) or a sleep transition from $\{(sleep, \tau, \iota), (leave, \tau, sleep)\}$ or a broadcast from the sequence generated from some $(l, b, A, l') \in t_E$, where $A = \{a_1, a_2, \dots, a_k\}$. A broadcast in γ is one of the following kind:

1. $(l, !(\langle a_1, \emptyset \rangle), b, l_1^e)$ and the corresponding receive transition in δ_C ,
2. $(l_{i-1}^e, !(\langle a_i, \{a_1, \dots, a_{i-1}\} \rangle), b, l_i^e)$ and the corresponding receive transition in δ_C ,
3. $(l_{k-1}^e, !(\langle a_k, \{a_1, \dots, a_{k-1}\} \rangle), b, l')$ and the corresponding receive transition in δ_C .

Note that, a sequence of broadcasts of a kind $(l, b, A, l') \in t_E$ in γ may be interleaved with sleep transitions but will never be interleaved with broadcasts of a different kind $(l^\dagger, b^\dagger, A^\dagger, l^\ddagger) \in t_E$. This is ensured by the construction of $\mathcal{R}_\mathcal{O}$. Thus, we can bunch together broadcasts of a kind by shifting all sleep transitions after a broadcast sequence is complete. This process will transform ρ to ρ' where there is no interleaving with sleep transitions.

Now, it is not difficult to see that we can derive a run $\varrho = g_0 \Longrightarrow g_1 \Longrightarrow \dots \Longrightarrow g_{\kappa'} = g$ of $\mathcal{O}$ from ρ' such that g is consistent with γ and therefore, we are done. □

Theorem 13. *The problem of safety checking in regular OMAS is P-time solvable.*

Proof. We appeal to Proposition 1 in [2] to decide the safety problem of regular OMAS in P-time. Given a regular OMAS $\mathcal{O}$ and a set of unsafe agent states $\mathbf{l}_u \subseteq L$, we encode $\mathcal{O}$ into $\mathcal{R}_\mathcal{O}$, using the encoding scheme given above, and compute a *cardinality constraint* $\varphi = \bigwedge_{l \in \mathbf{l}_u} (\#l \geq 1)$ in $CC[\geq 1]$. Then, using the P-time algorithm described in [2] we check whether φ holds in $\mathcal{R}_\mathcal{O}$. If the decision of the algorithm is yes then $\mathcal{O}$ is *unsafe* otherwise it is *safe*.

Additionally, we need to argue that the size of $\mathcal{R}_\mathcal{O}$ is within the prescribed limits. Here, we argue that size of $\mathcal{R}_\mathcal{O}$ is a quadratic order of the size of $\mathcal{O}$.

Recall, $|\mathcal{O}| = |L| \cdot |tr| + |L_E| \cdot |tr_E|$. Assuming $|L_E| = O(|L|)$ and $|tr_E| = O(|tr|)$ then, $|\mathcal{O}| = O(|L| \cdot |tr|)$. Further, we note that $|L|$ will be quite small compared to $|tr|$. So, we can safely assume $|\mathcal{O}| = O(|tr|^2)$.

$$|\mathcal{R}_\mathcal{O}| = |\mathcal{R}_C| + |\mathcal{R}_L| = |Q_C| \cdot |\delta_C| + |Q_L| \cdot |\delta_L|$$

From the encoding we can observe the following:

$$|Q_C| = O(|L|) \text{ and } |Q_L| = O(|L_E| + |tr_E| \cdot |Act|) = O(|tr_E| \cdot |Act|)$$

$$|\delta_C| = O(|tr| \cdot |tr_E|) \text{ and } |\delta_L| = O(|tr_E| \cdot |tr|)$$

So, $|\mathcal{R}_O| = O(|L| \cdot |tr| \cdot |tr_E|) + O(|tr_E| \cdot |Act| \cdot |tr_E| \cdot |tr|)$. Applying the above assumptions in the present context, we get $|\mathcal{R}_O| = O(|tr|^4)$.

□

5 Lower Bounds

6 Conclusion & Future Work

References

- [1] A. R. Balasubramanian and Chana Weil-Kennedy. Reconfigurable broadcast networks and asynchronous shared-memory systems are equivalent. In Pierre Ganty and Davide Bresolin, editors, *Proceedings 12th International Symposium on Games, Automata, Logics, and Formal Verification, GandALF 2021, Padua, Italy, 20-22 September 2021*, volume 346 of *EPTCS*, pages 18–34, 2021.
- [2] Giorgio Delzanno, Arnaud Sangnier, Riccardo Traverso, and Gianluigi Zavattaro. On the complexity of parameterized reachability in reconfigurable broadcast networks. In Deepak D’Souza, Telikepalli Kavitha, and Jaikumar Radhakrishnan, editors, *IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science, FSTTCS 2012, December 15-17, 2012, Hyderabad, India*, volume 18 of *LIPIcs*, pages 289–300. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2012.
- [3] Ronald Fagin, Joseph Y. Halpern, Yoram Moses, and Moshe Y. Vardi. *Reasoning About Knowledge*. MIT Press, 1995.
- [4] Panagiotis Kouvaros, Alessio Lomuscio, Edoardo Pirovano, and Hashan Punchihewa. Formal verification of open multi-agent systems. In Edith Elkind, Manuela Veloso, Noa Agmon, and Matthew E. Taylor, editors, *Proceedings of the 18th International Conference on Autonomous Agents and MultiAgent Systems, AAMAS ’19, Montreal, QC, Canada, May 13-17, 2019*, pages 179–187. International Foundation for Autonomous Agents and Multiagent Systems, 2019.
- [5] Shabana A. T., Alphy George, and S. Sheerazuddin. Counter abstraction for regular open teams. *Inf. Process. Lett.*, 188:106533, 2025.